

Switch Port Configuration – PBQ Answer Key with Explanations

Edited from the original key to show why each correct configuration is the expected answer.

VLAN key: 10 = WLAN, 20 = Server, 80 = Wired, 100 = Printers, 160 = Voice

How to read the explanations: Inter-switch links should be enabled as LACP trunks carrying all required VLANs. Endpoint ports should be enabled and assigned only the VLANs needed by the attached device(s). Unused ports should be administratively disabled for security and housekeeping.

Switch 1

Port(s)	Correct configuration	Why this is correct	Explanation
2, 4	Enabled, LACP, VLANs 10/20/80/100/160 (trunk)	Inter-switch uplink	These ports connect Switch 1 to Switch 2, so they must stay up, be bundled with LACP for redundancy/bandwidth, and carry every VLAN used across the network.
18, 26	Enabled, VLAN 80	Wired PCs	These ports feed PCs. Wired user endpoints belong on the Wired network, so each access port is enabled and placed in VLAN 80 only.
29, 33	Disabled	Unused ports	No active device is attached, so the safest and cleanest answer is to shut the ports down rather than leave unused access ports live.
38, 44	Enabled, VLAN 20	Servers	These ports connect to servers, so they should be active access ports in the Server VLAN only. They do not need LACP because each is a single host connection.

Switch 2

Port(s)	Correct configuration	Why this is correct	Explanation
1, 3, 37, 39, 47, 48	Enabled, LACP, VLANs 10/20/80/100/160 (trunk)	Core switch uplinks	Switch 2 is the transit point to the other switches. These are switch-to-switch links, so they stay enabled, use LACP where paired, and trunk all VLANs so traffic for every segment can cross the topology.
18, 28	Enabled, VLAN 20	Servers	Both ports serve servers, so they are simple enabled access ports in VLAN 20. A single server-facing edge port does not need to carry unrelated VLANs.

Switch 3

Port(s)	Correct configuration	Why this is correct	Explanation
1, 2	Enabled, LACP, VLANs 10/20/80/100/160 (trunk)	Uplink to Switch 2	These ports are the paired uplink back to Switch 2, so they are configured as an LACP trunk and must carry all network VLANs.
18, 28	Enabled, VLAN 80	Wired PCs	These are user workstation ports. The correct answer is an enabled access configuration in VLAN 80 for the wired client network.
34	Enabled, VLAN 100	Printer	This port connects to a printer, so it belongs in the printer segment only. That makes VLAN 100 the correct access VLAN.
39, 43	Disabled	Unused ports	Because there is no active endpoint shown on these ports, they should be disabled to reduce

			exposure and prevent accidental use.
48	Enabled, VLANs 10 and 80	AP + laptop/multi-network edge	This port supports multiple device/network needs, so it must carry both WLAN traffic (VLAN 10) and Wired traffic (VLAN 80). It remains a single active edge port, not an LACP bundle.

Switch 4

Port(s)	Correct configuration	Why this is correct	Explanation
1, 2	Enabled, LACP, VLANs 10/20/80/100/160 (trunk)	Uplink to Switch 2	These are the inter-switch uplinks back to Switch 2, so they should be enabled as an LACP trunk that carries every VLAN in use.
18	Enabled, VLANs 10, 80, 160	Multi-service endpoint	This edge connection supports more than one service type, so the correct answer includes the wireless/data/voice VLANs expected by the PBQ: VLAN 10, VLAN 80, and VLAN 160.
25, 33, 45	Disabled	Unused ports	Unused switch ports should be turned off. That prevents unauthorized connections and matches standard hardening practice.
32	Enabled, VLAN 100	Printer	The attached device is a printer, so this port is enabled as an access port in the printer VLAN only.
42	Enabled, VLAN 80	Wired PC	This is a standard wired client connection. The correct answer is an enabled access port in VLAN 80.

Pattern to remember: paired switch-to-switch links = LACP trunks with all required VLANs; single endpoint links = one or more task-specific access VLANs; unused links = disabled.